[Introduction to electronics] ECE[211s]

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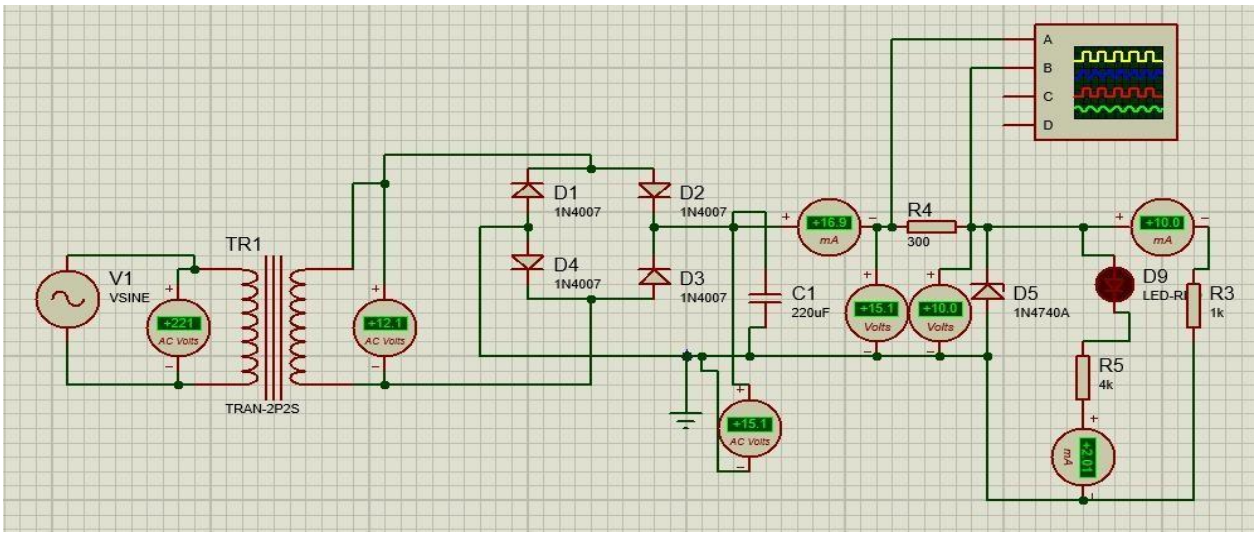
POWER SUPPLY

1- SCHEMATIC



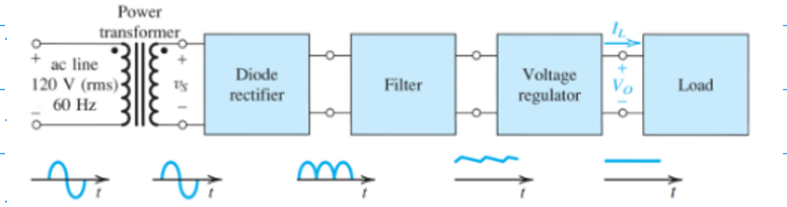
2- TABLE OF COMPONENTS

Component	Part no.	Price (EGP)	Role in circuit
Transformer 220V/12V		110	Converts 220v ac source to 12v ac source
Resistance (300 ohm)		0.25	Limit the current entering the Zener diode
Resistance (4.3 kohm)		0.25	Limit the current on the led
General Purpose Silicon Diode	1N4007	4*0.5=2	Building the bridge rectifier
capacitor (220uF) 25V		3	Smoothing ripples
Zener diode	1N4740A	0.5	Voltage regulation Overvoltage protection
Red led		3	Indicator of signal

3- HAND ANALYSIS

Electronics Project
Hand analysis

Part (a):



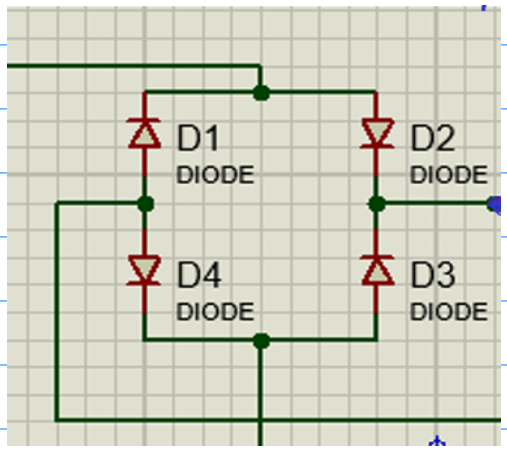
Voltage = 220v rms
frequency = 50 Hz

transformer is a step-down transformer (220 − 12).
∴ The output Voltage of the transformer is 12v rms

∴ $V_{out\ transformer}$ = 12v rms ≈ 17 V peak.

The Diodes used are model 1N4007

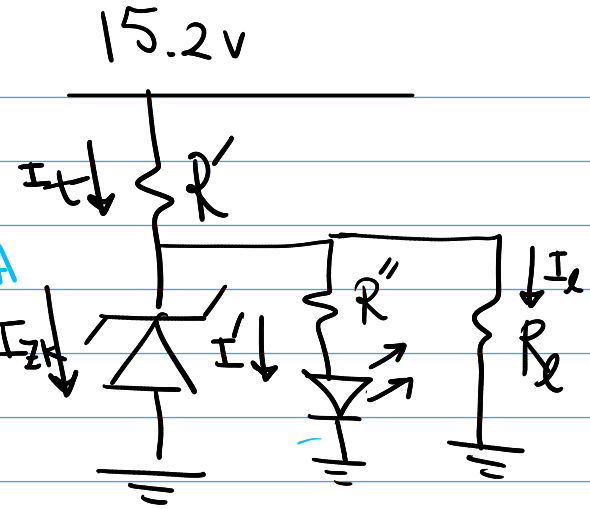
$V_{d,on} = 0.9V$



The Voltage for the output of the bridge rectifier:

$17 - 2(0.9) = 15.2$ V peak.

For the following part of the circuit



Worst case:

$\rightarrow I_l = 10mA \rightarrow I' = 2mA \rightarrow I_Z = 5mA$

$\therefore I_t = 10 + 2 + 5 = 17mA$

$\frac{15.2 - 10}{R'} = 17 \therefore R' = 306\,\Omega$

We will use a standard resistance of 300Ω

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for the LED branch:

$10 = 2\times10^{-3} R'' + V_{LED}$, $V_{LED} = 2.2v$ worst case

$\therefore 10 = 2\times10^{-3} R'' + 2.2 \quad \therefore R'' = 3900\,\Omega$

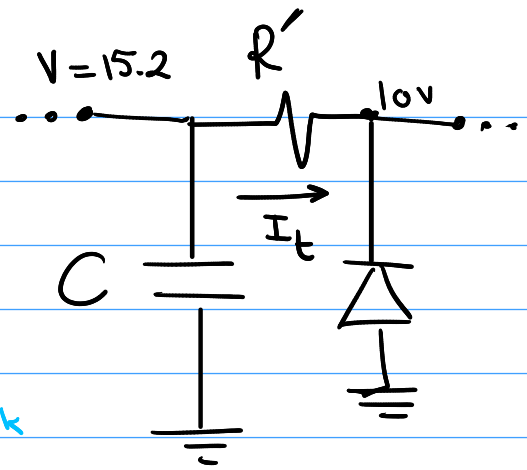
We will use a standard resistance of $4.3k\Omega$

For the load branch:

$10 = I_l R_l$, at maximum load current $= 10mA$:

$\therefore 10 = 10\times10^{-3} R_l \quad \therefore R_{l,min} = 1k\Omega$

For the filtering Capacitor:



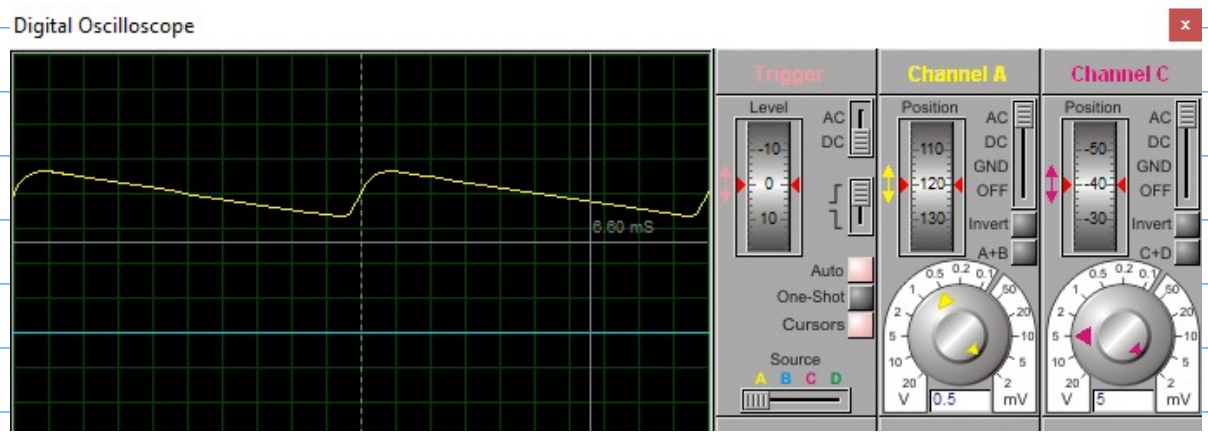
$$V_r = \frac{V_p}{2fRC} = \frac{i_t}{2fC}$$

for worst case, let $V_r = 1V$ peak-peak

$$\therefore C = \frac{i_t}{2fV_r} = \frac{17\times10^{-3}}{2(50)(1)} = 170\,\mu F$$

We will use a standard capacitance of $220\,\mu F$

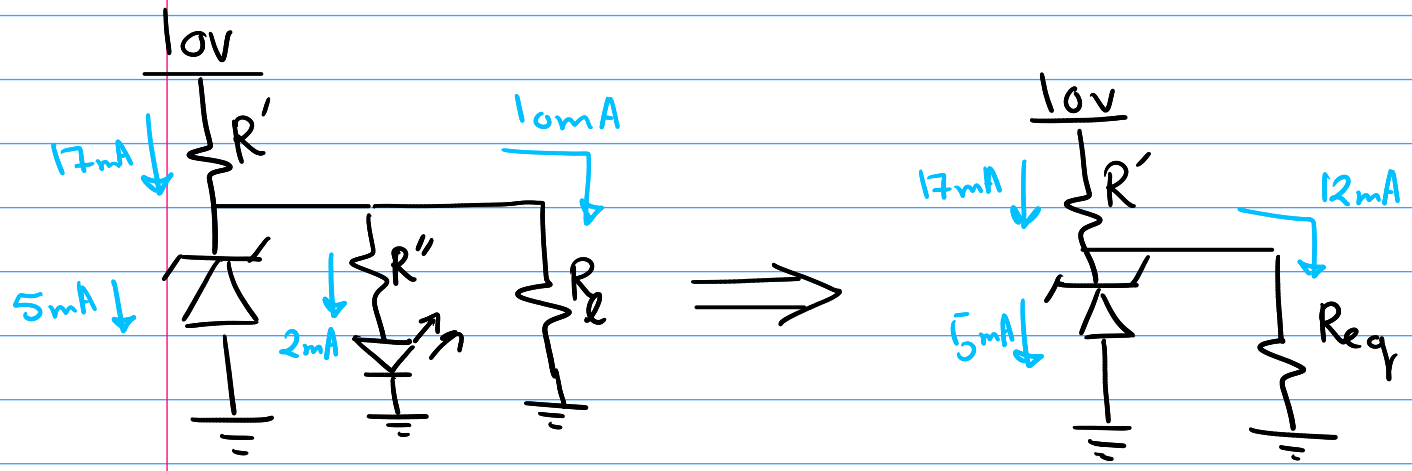
Therefore, $V_r = \dfrac{17\times10^{-3}}{2(50)\times220\times10^{-6}} = 0.77$ v peak-peak



the scale is 0.5v for every division.

For voltage regulation calculation:

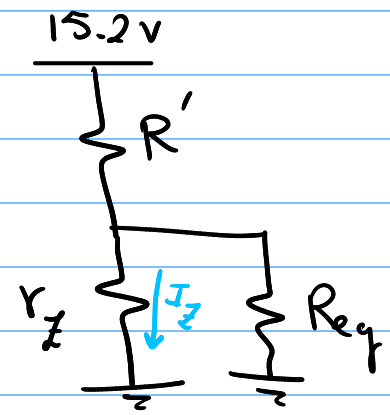
first we need to represent the load branch & the LED branch in an equivalent branch



$$10 = 12\times10^{-3}\, R_{eq} \quad \therefore R_{eq} = 833.33\,\Omega$$

Then, we need to do small signal analysis for the Zener diode:

$R_{eq} = 833.33\,\Omega$, $R' = 300\,\Omega$



$$V_z = V_{z0} + I_z r_z$$

→ from data sheet: $I_{z\,test} = 25\,mA \rightarrow r_z = 7\,\Omega$

$$\therefore V_{z0} = 9.825\ V$$

→ for our case:

$$10 = 9.825 + 5\times10^{-3}\, r_z \quad \therefore r_z = 35\,\Omega$$

Now, we can calculate voltage regulation:

$$\text{Voltage regulation} = \frac{\Delta V_{out}}{\Delta V_s}$$

$\rightarrow \Delta V_s = V_{r\,peak} = \frac{0.77}{2} = \pm 0.385\,V$ ((not peak to peak))

$$\Delta V_{out} = \pm 0.385 \left(\frac{r_z // R_{eq}}{300 + r_z // R_{eq}} \right) \text{, since } r_z << R_{eq}$$

$$\therefore \Delta V_{out} = \pm 0.385 \left(\frac{35}{300+35} \right) = \pm 40.22\,mV$$

$$\therefore \text{Voltage regulation} = \frac{+40.22 \times 10^{-3}}{0.385} = 104.47\,mV/V$$

Load Regulation:

$$\text{load regulation} = \frac{\Delta V_{out}}{\Delta i_{load}} = -r_z = -35\ mV/mA$$

4- PROS AND CONS USING BRIDGE RECTIFIER

Pros:

1- No need for a center-tapped transformer.
2- The PIV is one half that of the center-tap rectifier.
3- It can be used in applications allowing floating output terminals (no output terminal is needed to be grounded)

Cons:

1- Four diodes are required, which is more compared to other rectifiers (leads to more power loss).
2- Poor voltage regulation when low dc voltage is required.

5- PROS AND CONS USING ZENER DIODE

Pros:

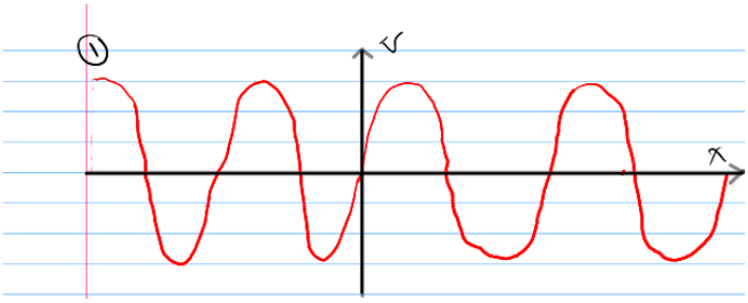
1- Zener diode is not expensive which makes this regulation economic
2- Simplicity in circuit design
3- It provides better regulation over a wider range of load currents and input voltages

Cons:

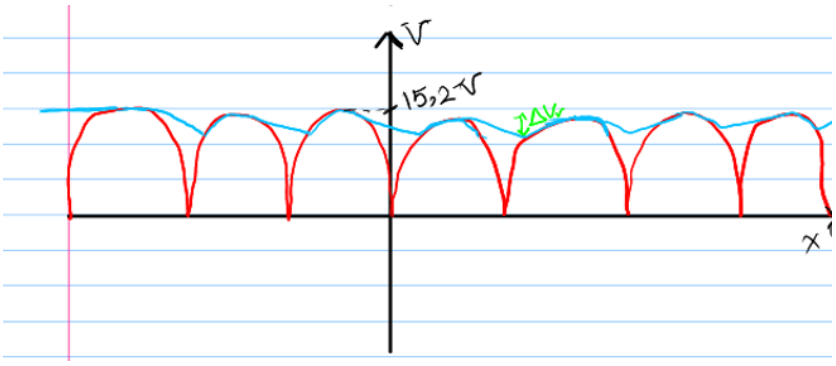
1- The DC output voltage slightly changes due to Zener resistance.
2- Cannot be used for high load current values as the Zener won't operate below certain current.
3- Poor efficiency as it constantly withdraws current whether with or without load.

6- WAVEFORM SKETCH

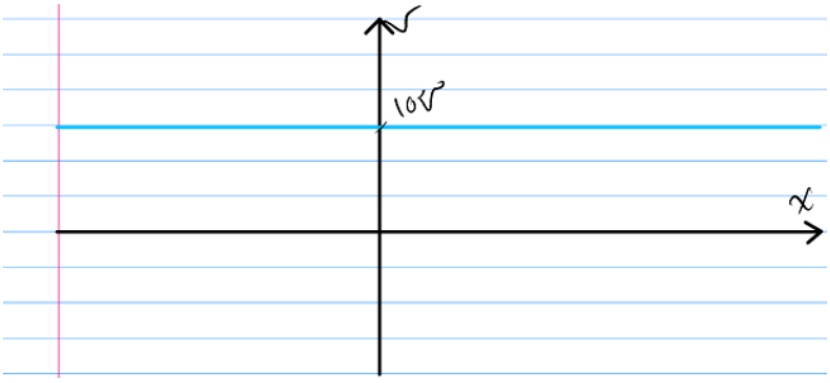
INPUT WAVEFORM:



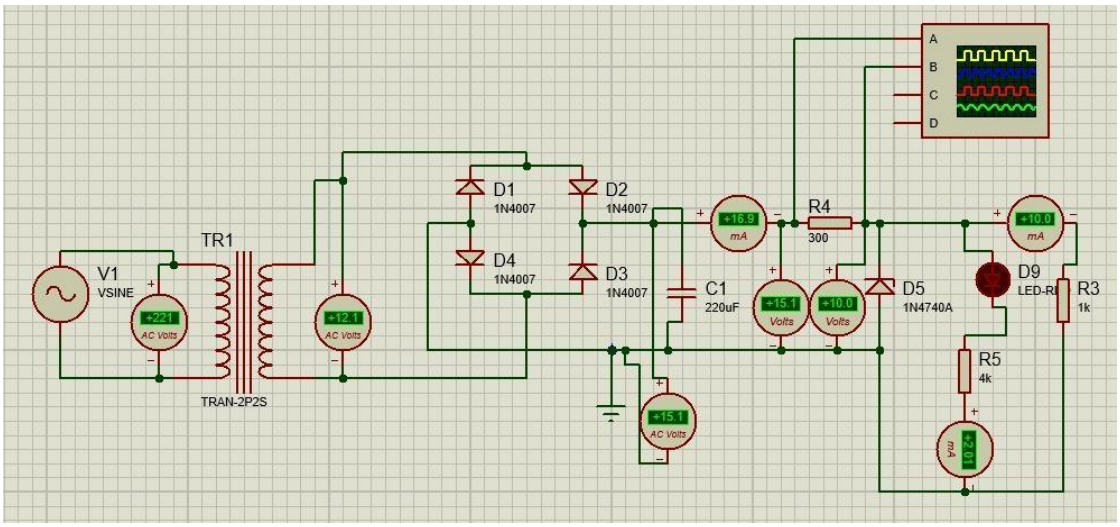
RECTIFED WAVEFORM:



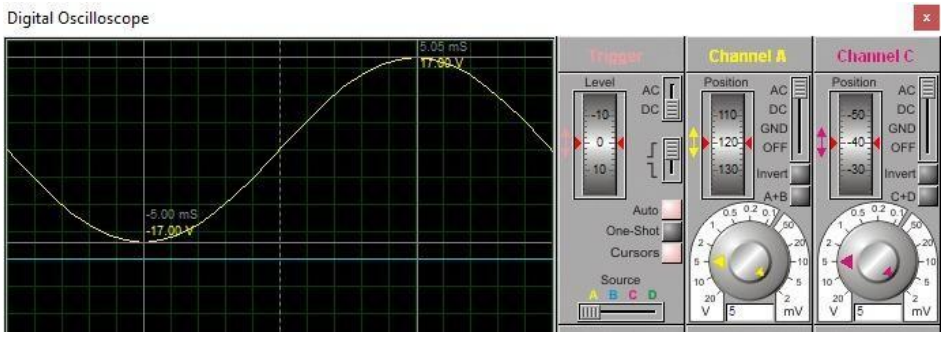
OUTPUT WAVEFORM:



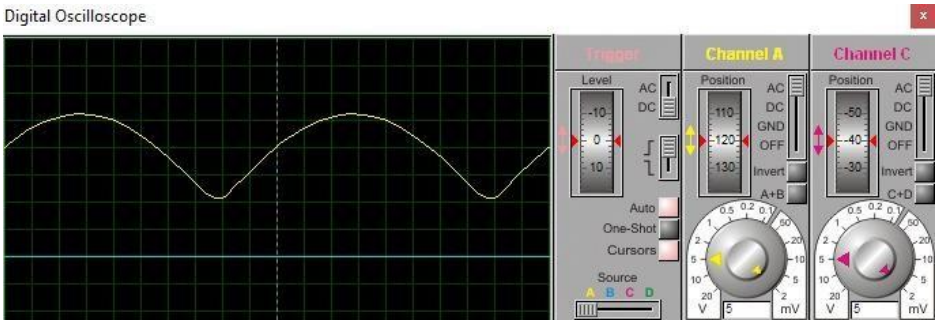
7- SIMULATION RESULTS



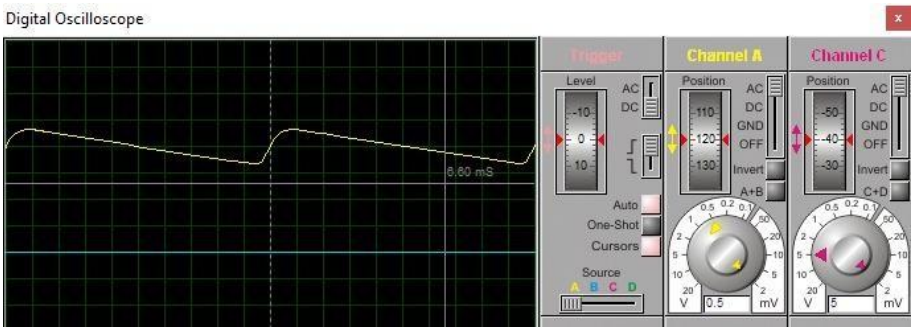
TRANSFORMER OUTPUT :



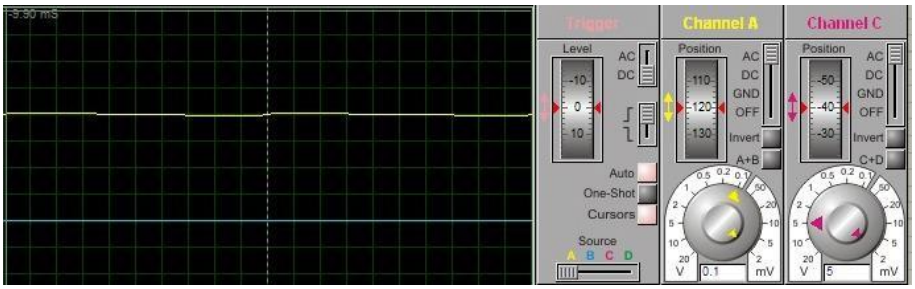
RECTIFIER OUTPUT WITHOUT CAPACITOR:



RECTIFIER OUTPUT WITH CAPACITOR:

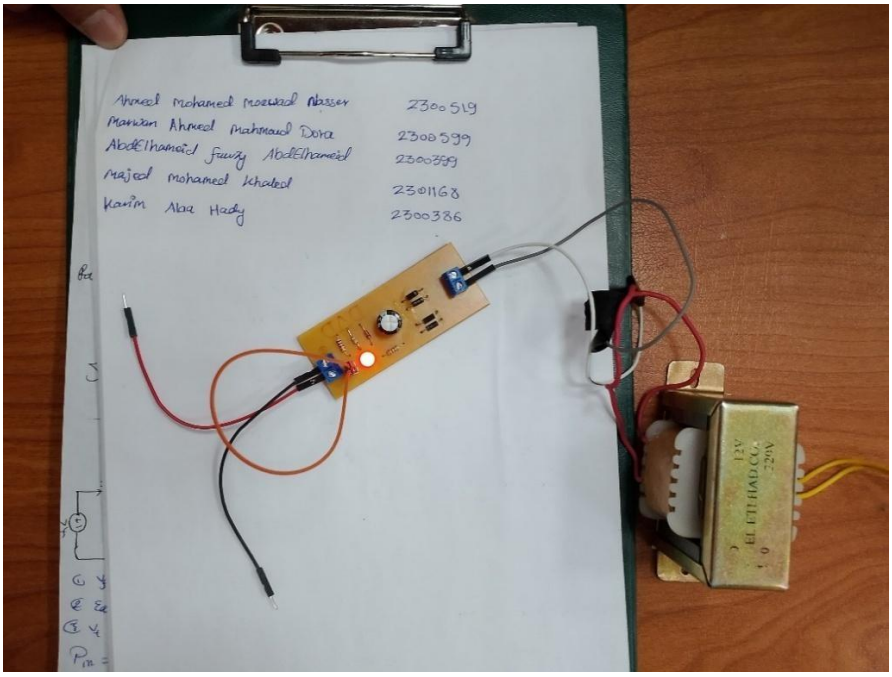


Regulator output:

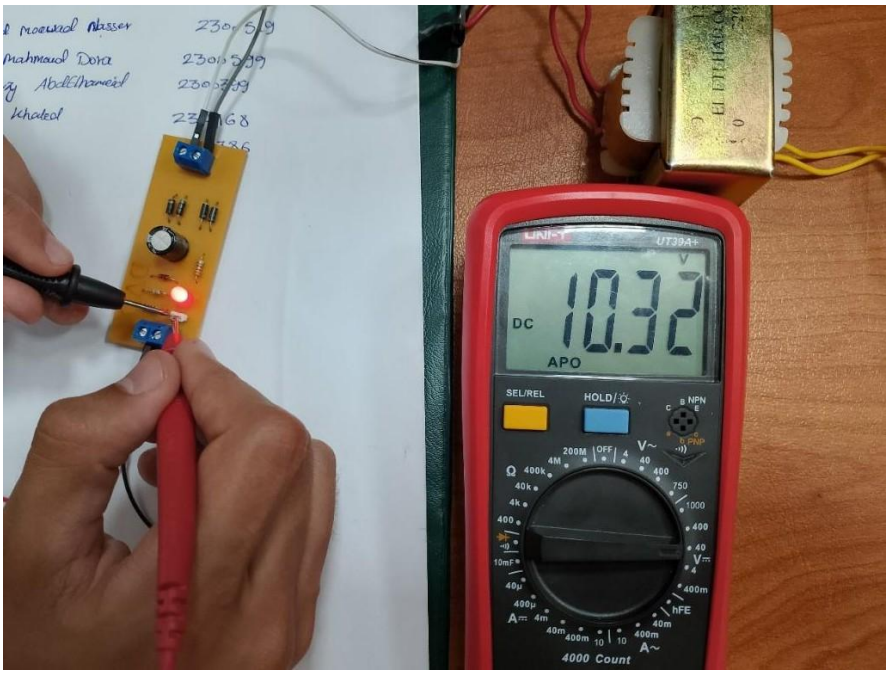


8- SNAPSHOTS OF THE PCB

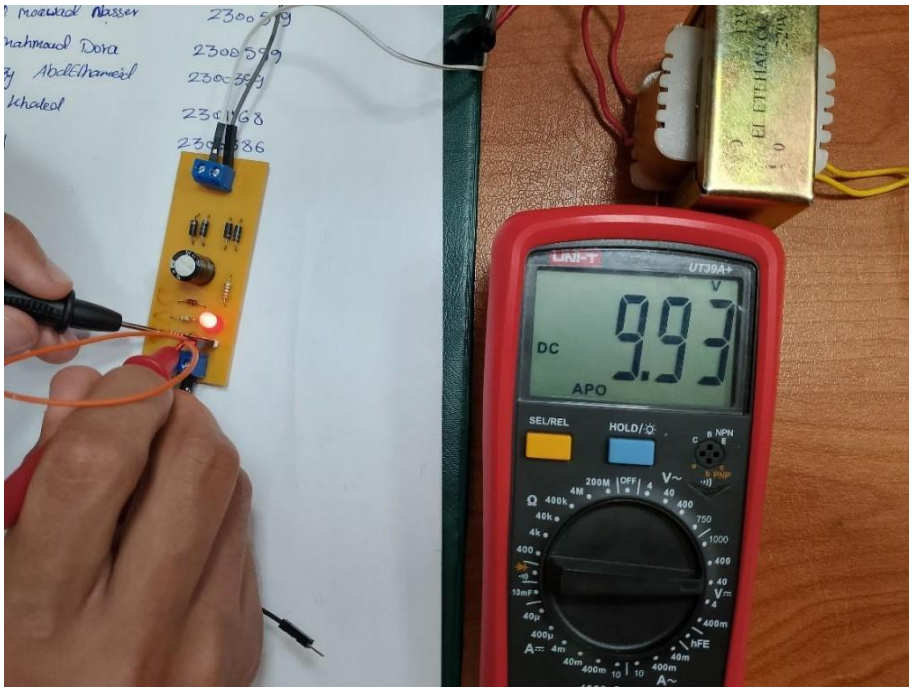
PCB OVERVIEW:



VOLTAGE WITH NO LOAD:



VOLTAGE WITH LOAD (1KOHM):



$$\therefore\ I_{load,max} = \frac{9.93}{1}\left(\frac{V}{Kohm}\right) = 9.93\ mA$$

9- GOOGLE DRIVE LINK

https://drive.google.com/drive/folders/1ZFdOjUp7VVTzSG0ZN4HTGeg6FB8Ka6iI

10- CONTRIBUTION TABLE

Name	Contribution
Ahmed	Simulation
Marwan	Simulation
Abdelhamid	Implementation
Karim	Implementation
Majed	Implementation

11- Refrences

http://www.polytechnichub.com

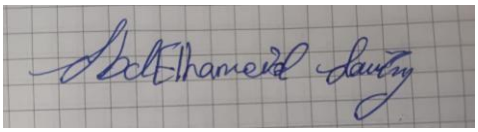
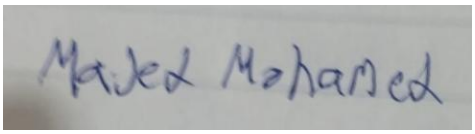
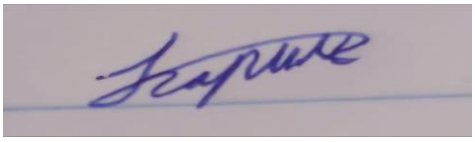
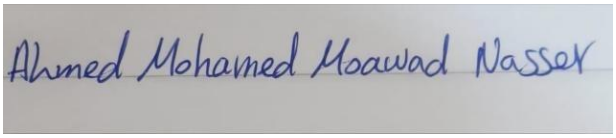
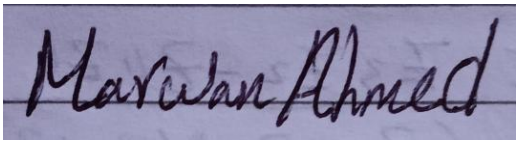
Zener diode datasheet:

1N4740A pdf, 1N4740A Description, 1N4740A Datasheet, 1N4740A view ::: ALLDATASHEET :::

Rectifier diodes datasheet:

1N4007 pdf, 1N4007 Description, 1N4007 Datasheet, 1N4007 view ::: ALLDATASHEET :::

Declaration:

“We certify that this report is our own work, based on our personal study and/or research and that we have acknowledged all material and sources used in its preparation, whether they are books, articles, reports, lecture notes, and any other kind of document, electronic or personal communication. We also certify that this assignment / report has not previously been submitted for assessment for another course. We certify that I have not copied in part or whole or otherwise plagiarized the work of other students and / or person s.”

12 – Signature